

Practice Quiz One Solution: KCL, KVL, and Ohm's Law (Easy)

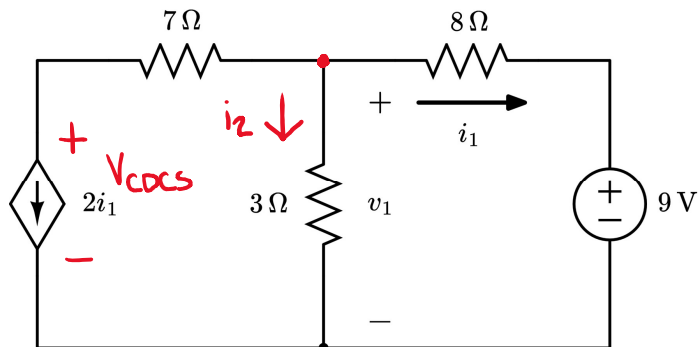
For the circuit shown below, find the following:

a) v_1

b) i_1

Finally, calculate the power of the current-dependent current source and state whether the power is absorbed or delivered.

Step One: KVL, right loop, following i_1 :

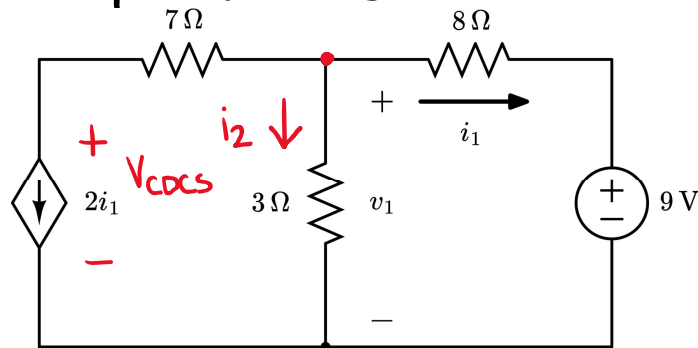


$$8i_1 + 9 - v_1 = 0$$

$$v_1 = i_2 \cdot 3 \quad \text{Ohm's Law}$$

$$8i_1 - 3i_2 = -9$$

Step Two: KCL



KCL at essential node:

$$i_1 + 2i_1 + i_2 = 0$$

$$i_2 = -3i_1$$

Step Three: Solve system of equations

$$8i_1 - 3i_2 = -9$$

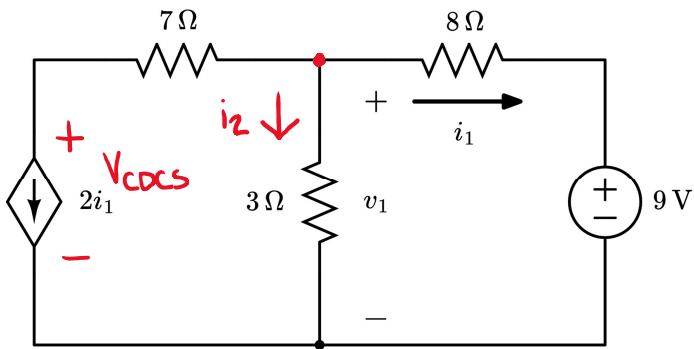
$$8i_1 - 3(-3i_1) = -9 \quad \text{Substitute } i_2 = -3i_1$$

$$i_1 = -\frac{9}{17} \text{ A}$$

$$i_2 = -3i_1 = \frac{27}{17} \text{ A}$$

$$v_1 = i_2 \cdot 3\Omega = \frac{81}{17} \text{ V}$$

Step Four: KVL, outside loop:



$$7(-2i_1) + 8i_1 + 9 - v_{CDCS} = 0$$

$$v_{CDCS} = \frac{207}{17} \text{ V}$$

Step Five: Power Calculation

$$P_{CDCS} = v_{CDCS} \cdot 2i_1 = \frac{207}{17} \text{ V} \cdot 2 \cdot -\frac{9}{17} \text{ A}$$

$$P_{CDCS} = -12.89 \text{ W} \text{ Delivering}$$

Self-Grading Rubric	
Used Ohm's law to make a relationship with v_1	+2 points
Correct value for v_1	+2 points
Correct value for i_1	+2 points
Correct sign for the power of the current-dependent current source	+2 points
Correct magnitude for the power of the current-dependent current source	+2 points